

DMA Controller

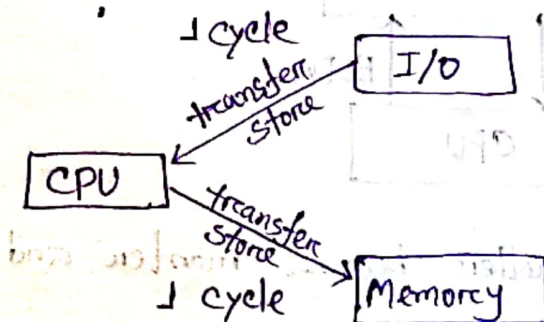
DMA — Direct Memory Access

Definition: Transferring data directly between memory and I/O without any involvement of microprocessor at fastest rate.

Bus master — controls the system bus (eg- microprocessor)

** Since in DMA, there's no involvement of microprocessor, someone has to be the bus master. But neither memory nor I/O can be the bus master, so we use DMA controller as the bus master.

** যখন microprocessor transfer করে তখন 2 cycles complete করে
এই সময় বেশি লাগে।



↳ fetching and executing instruction

but DMA controller just address & control signal generate করে
direct transfer করে এই সময় কম লাগে (1 cycle)

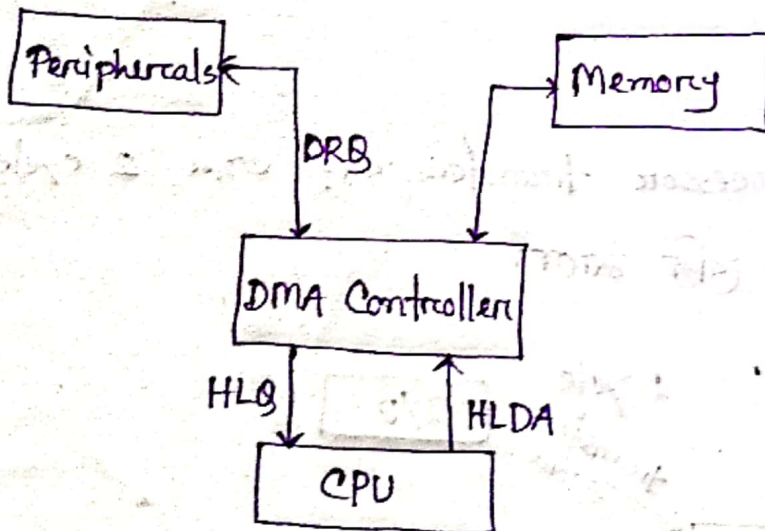
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* How DMA operations are performed?

- ① If the device (peripheral) wants to send data to memory, first device has to send DMA request (DRQ) to DMA controller.
- ② DMA controller send HLQ (hold request) to CPU (to remain hold and hand over the system bus to DMA controller) & waits for the CPU to send HLDA (handing over the system bus to DMA controller).
- ③ The CPU leaves the control over bus and acknowledge the HLDA signal to DMA controller.
- ④ Now CPU is in hold state, DMA controller has to manage operations over buses between CPU, memory, I/O devices.



DMA Controller becomes master and CPU becomes slave.

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* DMA Controller Modes for data transfer:

1. Burst mode: Bus को control करता होता है जब transfer complete हो जाता है तब bus को control CPU को मिलेगा।
2. Cycle stealing mode: DMA forcefully bus को control लिए 1 byte transfer करेगा, तब CPU को control मिलेगा, तब CPU 1 byte transfer को पूरा bus को control लेगा, इस CPU को request करेगा।
3. Transparent mode: CPU जब Idle होगा तब bus को control लिए transfer करेगा।

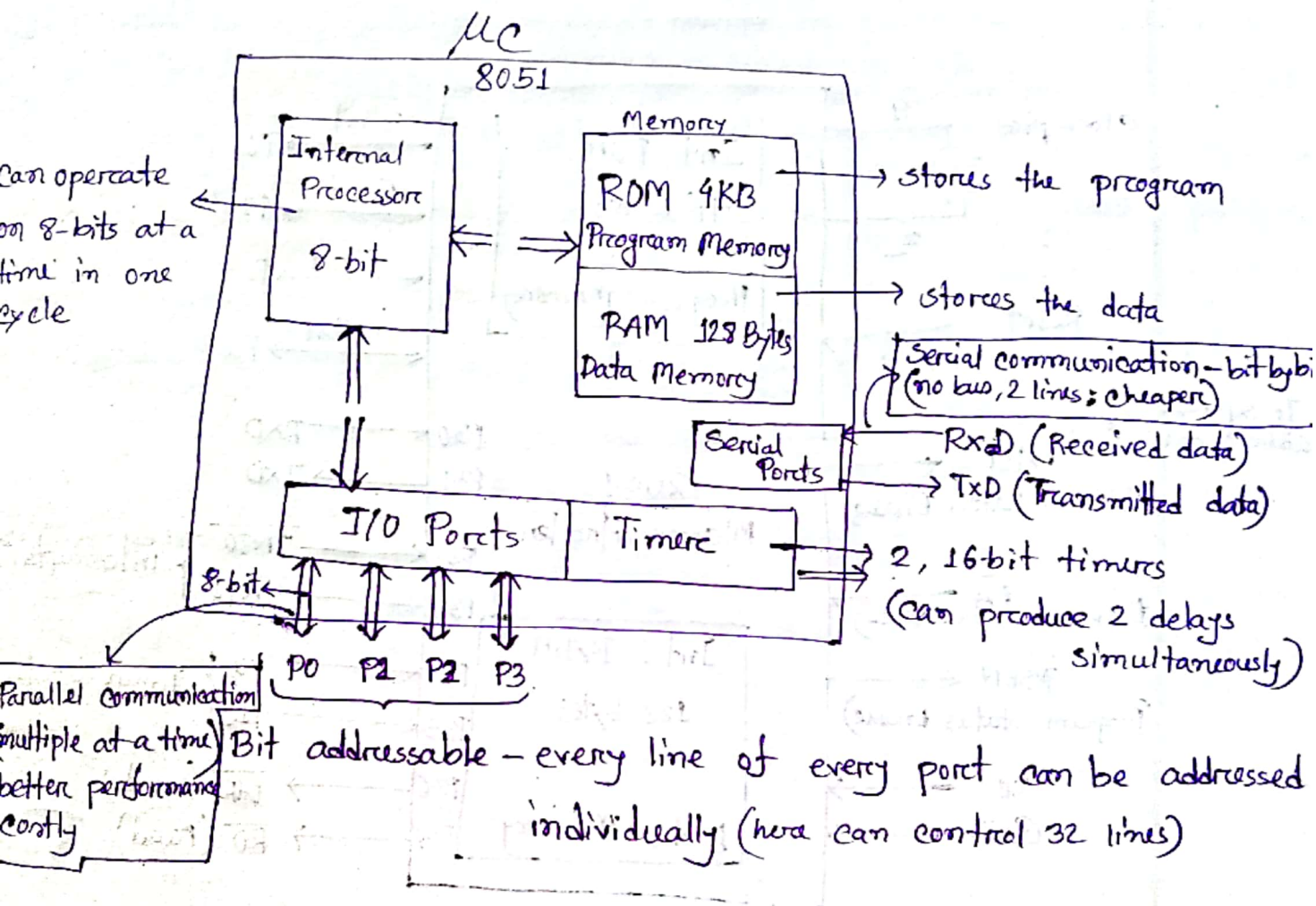
Hardware delay - timer produces delay, processor performs other operations
 Software delay - processor produces delay

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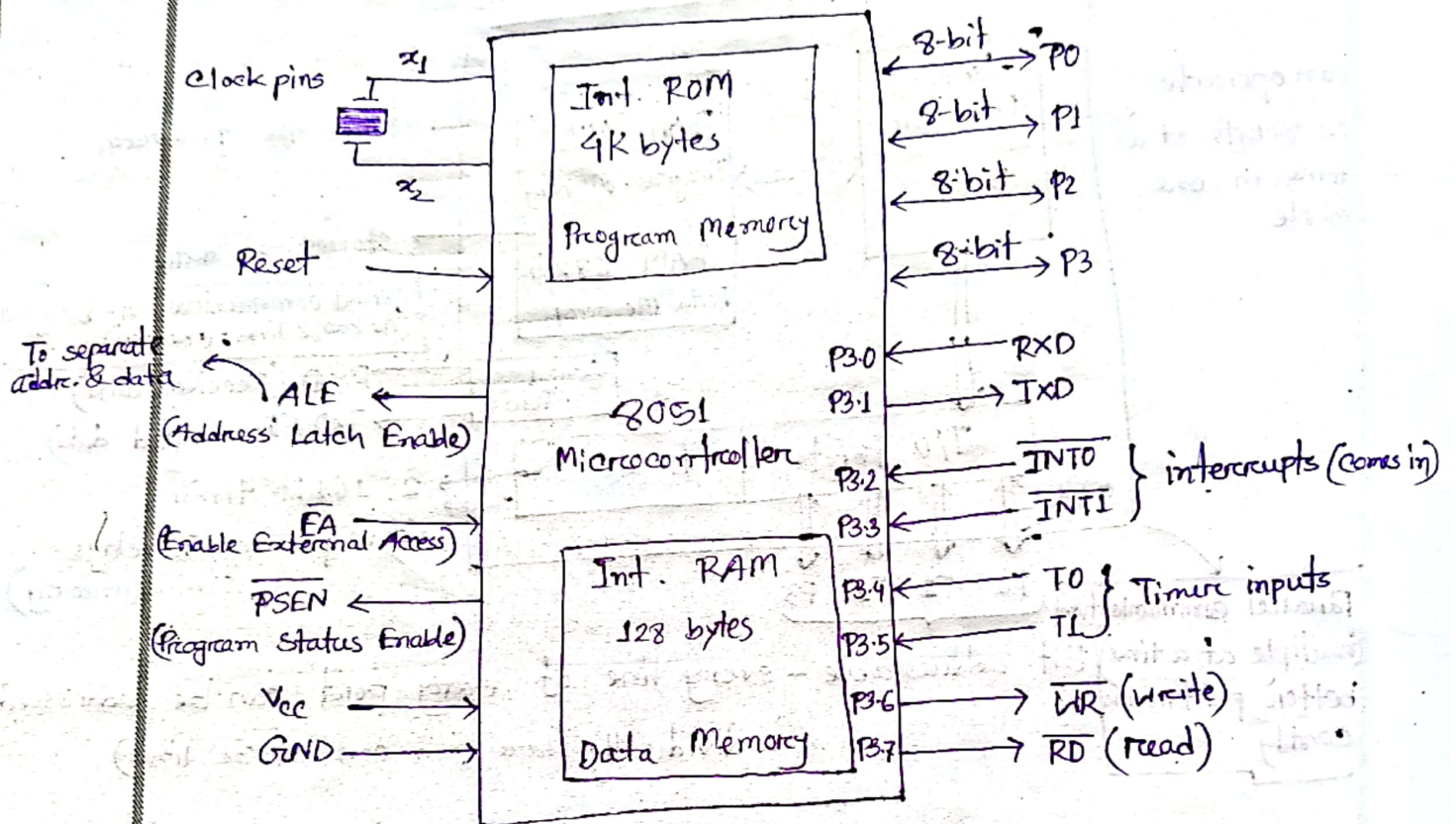
8051 Microcontroller



Microcontroller: A microcontroller is a compact integrated circuit designed to govern a specific operation in an embedded system. A typical microcontroller includes a processor, memory, i/o peripherals on a single chip.

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* 8051 Pin Diagram



United

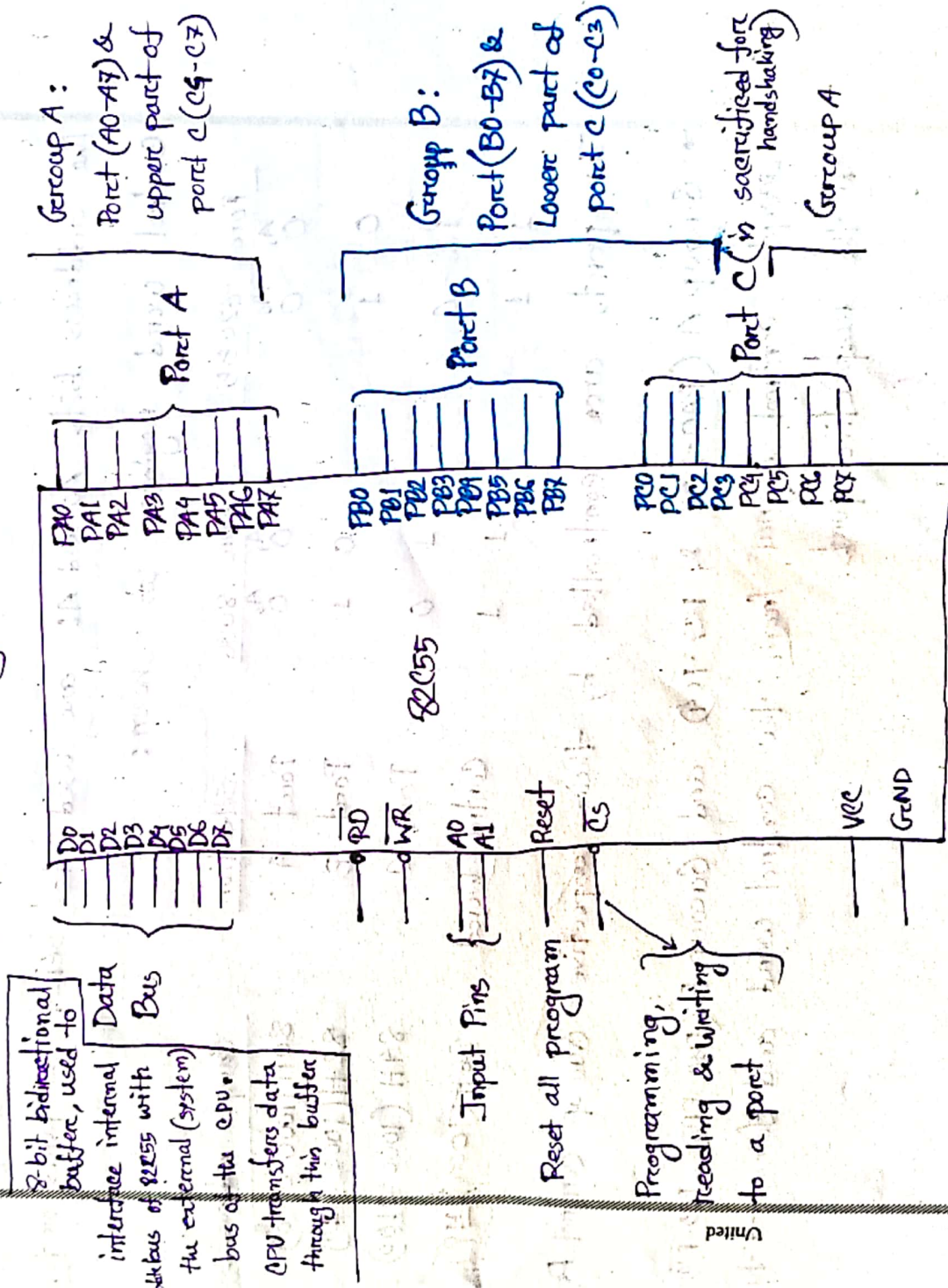
** Feature	8051	8052	8031 (Rom-less)
ROM (on chip)	4K	8K	0K
RAM (bytes)	128	256	128
Timers	2	3	2
I/O Pins	32	32	32
Serial port	1	1	1
Interrupt sources	6	8	6

PPI

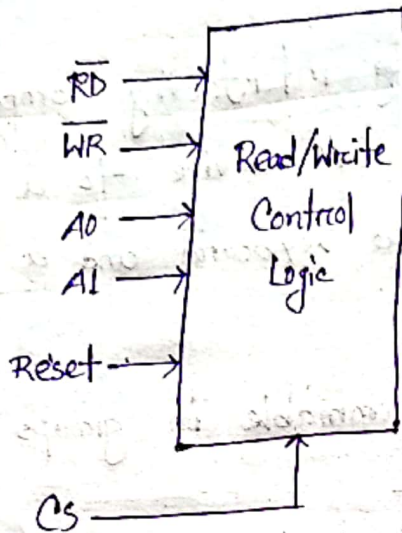
*82C55 is a popular low cost interfacing component, that can interface any TTL-compatible I/O device to a microprocessor. It is used to interface to the keyboard and a parallel printer port in PCs.

- 24 pins for I/O that are programmable in groups of 12 pins.

Pin Layout



Control Signals: (sends commands)



accepts address and control signals from the μP .

The control signal determines whether it is a read or a write operation also select or reset the 82C55 chip.

The address bits A_0 and A_1 are used to select the ports or the control word register as shown:

For 82C55		For 8086		Selection	Sample address
A_1	A_0	A_1	A_0		
0	0	0	0	Port A	80H (1000 0000)
0	1	0	1	Port B	82H (1000 0010)
1	0	1	0	Port C	84H (1000 0100)
1	1	1	1	Control Word	86H (1000 0110) bank

The ports are controlled by their respective Group Control Reg.

* Group A (PA_0 - PA_7 & PC_7 - PC_4) and Group B (PB_0 - PB_7 , PC_3 - PC_0) accepts control signal from the control word and forwarded them to the respective ports.

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* Port A, B, C are bidirectional 8-bit Ports. They can be programmed to work in the various modes as follows:

Port	(Simple I/O) Mode 0	(Handshaking I/O) Mode 1	(Bidirectional handshaking) Mode 2
Port A	Yes	Yes	Yes
Port B	Yes	Yes	No (Mode 0 or 1)
Port C	Yes	No (Handshake signals)	No (handshake signals)

* Only port C can also be programmed to work in Bit set reset Mode to manipulate its individual bits.

I/O Mode/I/O command

Command Byte A (Programs Port A, B, C)

7 6 5 4 3 2 1 0
D7 D6 D5 D4 D3 D2 D1 D0

Mode Set Flag
1 = Active

Selects the mode of port A

Selects direction of Port A

Direction of Port C (lower)

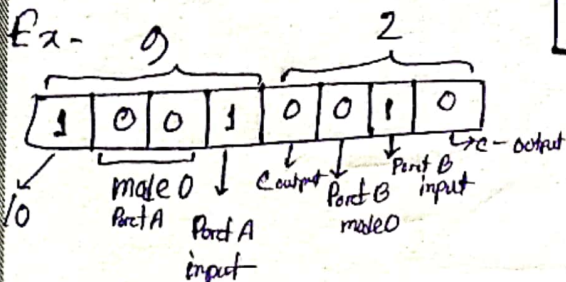
Direction of Port B

mode of Port B

Direction of Port C (upper)

Group B
Port C (Lower)
1 = Input
0 = Output
Port B
1 = Input
0 = Output
Mode Selection
0 = Mode 0
1 = Mode 1

Group A
Port C (Upper)
1 = Input
0 = Output
Port A
1 = Input
0 = Output
Mode Selection
00 = Mode 0
01 = Mode 1
1x = Mode 2



92H

MOV AL, 92H

OUT 86H, AL

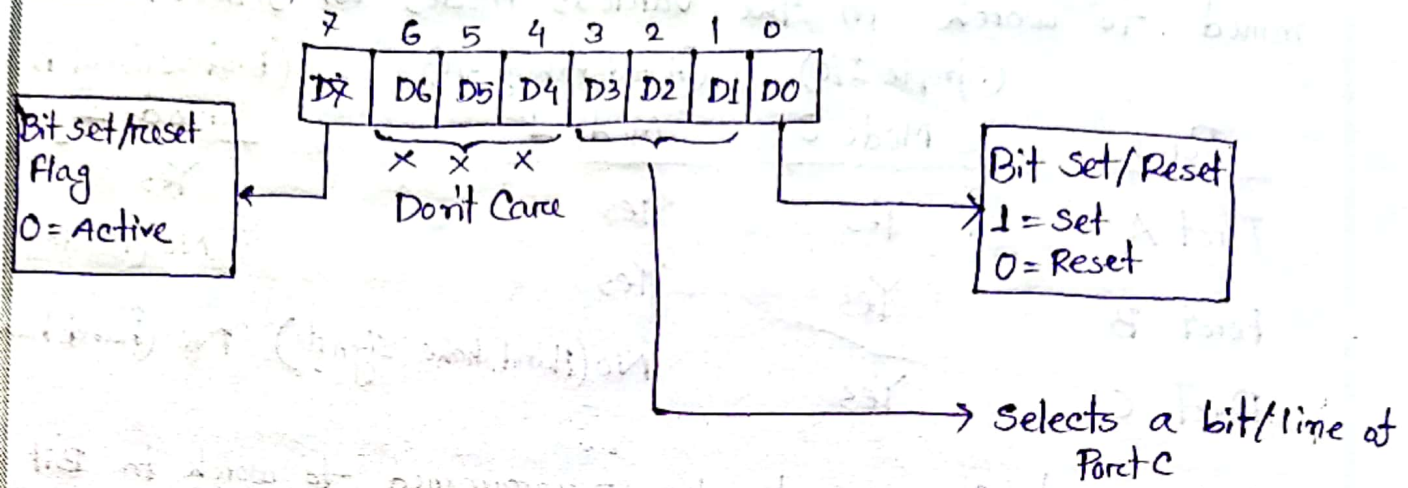
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Reset command

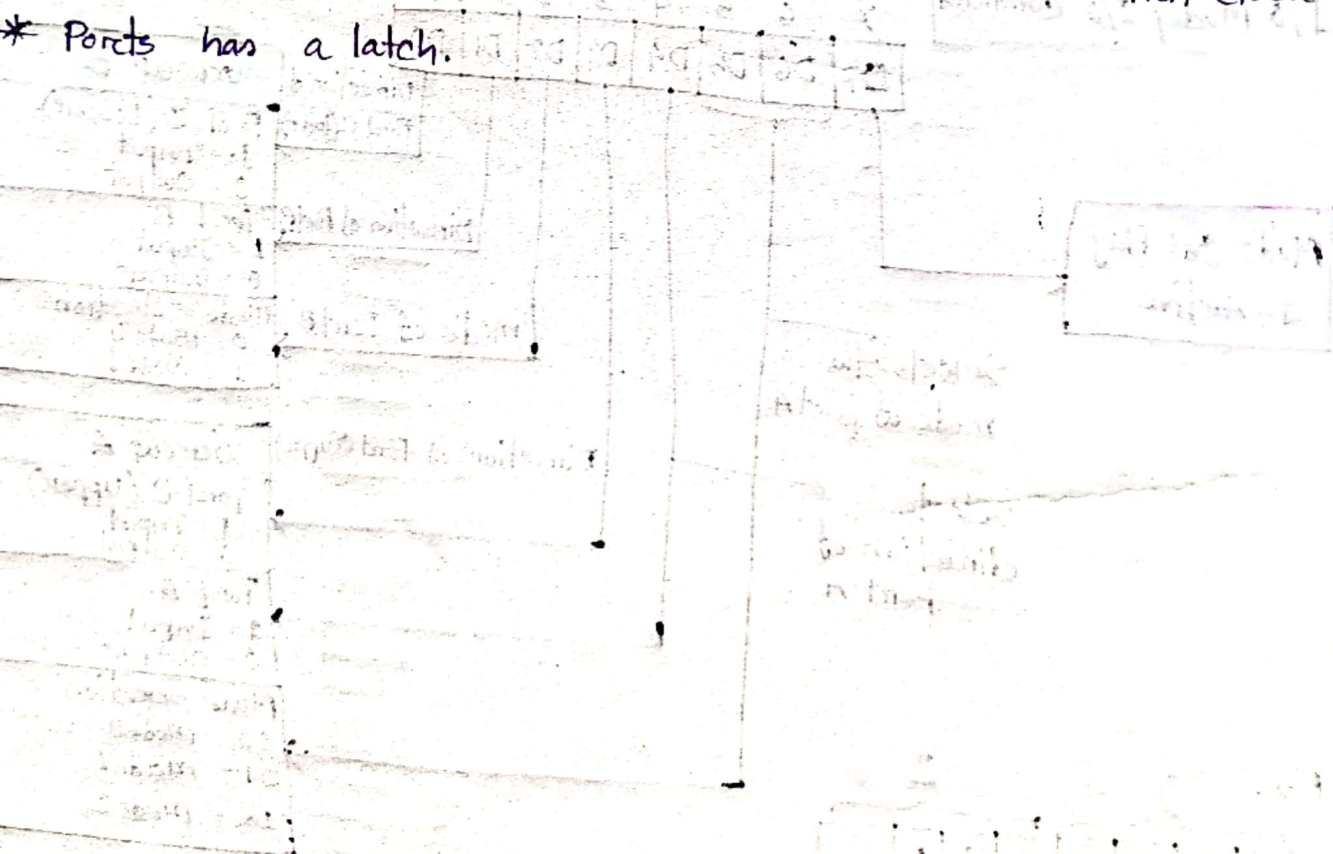
Command Byte B (Sets or resets any bits in Port C)



* We use 82C55 as it can perform handshaking.

* 82C55 doesn't care about time so there's no clock signal

* Ports has a latch.



Memory Interfacing

Memory interfacing: The interfacing process includes some key factors to match with the memory requirements and microprocessor signals. The interfacing circuit therefore should be designed in such a way that it matches the memory signal requirements with the signals of the μP .

→ Memory interfacing is needed/used to access memory quite frequently to read instruction codes and data stored in memory. This read/write operations are monitored by control signals. The μP activates these signals when it wants to read from and write into memory.